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Bulk tissue transcriptomes obscure the IDI2-AS1 / IL5 relationship through cell-composition effects across four type-2 allergic diseases

Hidenori Tani

Abstract

Background: The long noncoding RNA *IDI2-AS1* was recently identified in vitro as a negative regulator of interleukin 5 (*IL5*), a target of approved anti-eosinophil biologics, motivating an in vivo test in *IL5*-driven allergic disease.

Methods: We re-analyzed five public bulk RNA-seq cohorts ($n = 856$) covering asthma (GSE152004), atopic dermatitis (AD; GSE121212), eosinophilic esophagitis (EoE; GSE246323; GSE58640), and chronic rhinosinusitis with nasal polyposis (CRSwNP; GSE136825) under a uniform DESeq2/limma pipeline and pooled by random-effects meta-analysis. Sample-level *IDI2-AS1* ↔ *IL5* Spearman correlations were re-estimated as partial correlations conditioning on eosinophil and Th2 marker-signature scores. In

the largest cohort, a bootstrap mediation decomposition split the association into direct (ADE) and mediated (ACME) components.

Results: Bulk-tissue *ID12-AS1* was invariant in every cohort (pooled $\log_2\text{FC} = +5.1 \times 10^{-5}$; $I^2 = 0.26\%$; $p = 0.47$). Sample-level *ID12-AS1* and *IL5* covaried *positively* in every testable cohort, significantly so in asthma ($\rho = +0.109$, $p = 0.004$) — opposite to the in vitro repressive direction. Conditioning on the eosinophil signature attenuated the asthma association to $\rho = +0.052$. The asthma-cohort decomposition attributed $\approx 90\%$ of the covariation to the eosinophil compartment (ACME = $+0.109$, $p = 0.002$), with a direct effect indistinguishable from zero (ADE = $+0.012$, $p = 0.76$). Comparator lncRNAs and canonical type-2 cytokines showed the expected disease-stratified pattern.

Conclusions: Bulk patient transcriptomes do not recover the in vitro repressive direction. The pattern is compatible with masking of any per-cell *ID12-AS1* $\rightarrow$ *IL5* effect by eosinophil-composition shifts, although we cannot exclude the alternative that the in vitro repression does not generalize to patient cells; both possibilities require cell-type-resolved follow-up for adjudication. Beyond the *ID12-AS1* case, the cell-composition-adjusted re-analysis pipeline described here provides a reusable framework for interpreting bulk-tissue null results for other low-abundance, cell-type-restricted lncRNA candidates.

Keywords: Asthma; Cell composition; Eosinophil; IL5; Long noncoding RNA.

Introduction

Long noncoding RNAs (lncRNAs) constitute the largest class of transcribed elements in the human genome and have emerged in the past decade as a distinct layer of post-transcriptional and transcriptional regulation, with roles documented in development, immune cell differentiation, and disease [1,2]. Compared with protein-coding mRNAs, lncRNAs are characteristically lower in steady-state abundance, often short-lived [3], and more cell-type-restricted in their expression [4]. These properties have practical consequences for translational lncRNA research: a regulator whose biological function is concentrated in a small effector subset may be functionally important in vivo while being effectively undetectable in standard bulk-tissue transcriptomics, where its per-cell contribution is averaged across thousands of bystander cells. This implies that standard bulk-tissue RNA-seq catalogs may systematically under-report lncRNAs whose activity is confined to rare immune effector subsets.

We recently reported that *ID12-AS1*, an antisense lncRNA transcribed from the *ID12* locus, modulates expression of the type-2 effector cytokine *IL5* in human cells [5]. Using siRNA-mediated knock-down in HuT78 T-cells and orthogonal validation in

A549 epithelial cells, we showed that loss of *IDI2-AS1* increases steady-state *IL5* mRNA, identifying the lncRNA as a candidate negative regulator of *IL5* expression. Although several other lncRNAs have been implicated in type-2 / allergic inflammation, *IDI2-AS1* is mechanistically distinct in that it directly targets *IL5*, the most clinically tractable type-2 cytokine.

The *IL5* axis is one of the rare success stories of molecularly targeted therapy in allergic disease, with multiple anti-*IL5* / anti-*IL5Rα* biologics now in clinical use ([6]; detailed clinical context is expanded in the Discussion). The clinical efficacy of these protein-level interventions establishes that *IL5* is dose-limiting for the eosinophilic phenotype in vivo, which in turn motivates a search for upstream regulators — including non-coding regulators — that might extend the therapeutic landscape, particularly in directions where the cell-type-specificity of an oligonucleotide-based modulator could complement the systemic effects of an antibody.

A natural next step after our in vitro report was therefore to ask whether the *IDI2-AS1* / *IL5* axis is detectable, and operates in the predicted direction, in patient tissue from human *IL5*-driven allergic diseases. Public bulk RNA-seq cohorts now span the four major type-2 inflammatory diseases — asthma [7], atopic dermatitis [8], eosinophilic esophagitis [9,10], and chronic rhinosinusitis with nasal polypsis [11] — at sample sizes that, in the largest cohort, exceed several hundred patients. Here, we re-analyzed five public bulk RNA-seq datasets under a uniform pipeline (Fig. 1) to test three in vivo questions about the in vitro mechanism: (i) is *IDI2-AS1* differentially expressed between patients and controls in any of these tissues; (ii) does sample-level *IDI2-AS1* abundance correlate with *IL5* abundance in the direction predicted by the in vitro repression model (negative); and (iii) if the bulk-tissue signal departs from the in vitro prediction, can any of it be attributed to eosinophil- and Th2-associated shifts in tissue cell composition? We hypothesized that any per-cell repressive *IDI2-AS1* → *IL5* effect operating in allergic effector cells would be largely masked in bulk tissue by disease-associated changes in eosinophil and Th2 cell abundance, and designed the analysis to separate these two contributions. In addressing these questions, the present study also offers a reusable cell-composition-adjusted re-analysis framework applicable to other low-abundance, cell-type-restricted lncRNA candidates evaluated against bulk-tissue cohorts.

Methods

Ethics

This study used only publicly available, fully de-identified RNA-seq datasets that were obtained with appropriate ethical approval by the original investigators. No new

human or animal data were generated, and no additional ethical approval was required for the present re-analysis.

Dataset selection and acquisition

Public bulk-tissue RNA-seq datasets from human IL5-driven allergic diseases were identified by querying the NCBI Gene Expression Omnibus (GEO) and ArrayExpress in March 2026. Inclusion criteria were: (i) human disease tissue with paired or matched controls; (ii) RNA-seq (not microarray) at sufficient depth to detect long noncoding RNAs; (iii) availability of per-sample expression matrices (raw counts or FPKM) — datasets providing only pre-computed differential-expression tables were excluded; and (iv) detection of *ID12-AS1* (ENSG00000260196) as a non-zero feature in the supplementary expression matrix. After applying these criteria, five datasets covering four allergic diseases were retained for analysis (Table 1): asthma (GSE152004, $n = 695$ nasal epithelial brushings; [7]); atopic dermatitis (GSE121212, $n = 65$ skin biopsies; [8]); eosinophilic esophagitis (GSE246323, $n = 10$ esophageal biopsies, [10]; and GSE58640, $n = 16$ esophageal biopsies, [9]); and chronic rhinosinusitis with nasal polyposis (GSE136825, $n = 70$ nasal polyp and turbinate biopsies; [11]). Three additional candidate datasets (GSE201955, GSE65832, GSE179269) were excluded because *ID12-AS1* was absent from the available processed matrix or because per-sample raw data could not be recovered without re-quantification from SRA. For GSE136825, only the patient nasal-polyp biopsies (NP, $n = 42$) and unrelated healthy-donor inferior-turbinate biopsies (Control IT, $n = 28$) were used; the within-patient paired turbinate samples (NP_IT, $n = 33$) were excluded so that case/control contrasts in this cohort were structurally comparable to those in the other four datasets. Sample-to-condition mapping was reconstructed for every dataset from the GEO series matrix metadata; assignments are documented in `data/metadata/final_datasets.csv` and the per-dataset loader scripts (`code/R/loaders/load_GSE*.R`). We acknowledge that inclusion criterion (iv) restricts the conclusions of this study to cohorts in which *ID12-AS1* is detectably expressed at the bulk level, and that datasets in which the transcript is absent from the processed matrix (whether for biological or technical reasons) are by construction not represented.

Differential expression analysis

For the four raw-count datasets (GSE152004, GSE121212, GSE246323, GSE136825), low-count features (rowSum < 10 across the dataset, a conventional DESeq2 pre-filter used to remove genes with extremely sparse counts that are unlikely to yield stable fold-change estimates) were removed and differential expression was computed with DESeq2 (v1.42.1; [12]) under a single-factor design (~ group, control as reference) using the Wald test with Benjamini–Hochberg FDR correction. Effect

sizes for the patient-vs-control contrast were stabilized by adaptive shrinkage with `apeglm` ([13]; `lfcShrink(coef = "group_patient_vs_control", type = "apeglm")`), and the resulting log2 fold-changes and posterior standard errors were used as the input statistics for downstream meta-analysis. For the FPKM dataset (GSE58640), expression was $\log_2(\text{FPKM} + 1)$ -transformed and analyzed with `limma` (v3.58.1; [14]) using the `limma-trend` workflow (`lmFit` followed by `eBayes(trend = TRUE)`); coefficient standard errors were recovered as `stdev.unscaled[, 2] * sigma`. A focal-gene panel covering the lncRNA of interest (*IDI2-AS1*, ENSG00000260196), four comparator lncRNAs (*MIR22HG*, *GABPB1-AS1*, *OIP5-AS1*, *LITATS1* — selected because the same lncRNAs were previously identified by our group as bacterial- and viral-stress-responsive in human cells; [15]; [16]), and six type-2 / inflammatory cytokines (*IL5*, *IL4*, *IL13*, *IL6*, *TNF*, *IFNG*) was extracted from each per-dataset table for figure construction; the full per-dataset DEG tables are provided in `results/tables/`. Of the four comparator lncRNAs, *LITATS1* fell below the rowSum count threshold in all symbol-keyed allergic-tissue datasets and is therefore reported only at the level of dataset-wise availability and not in the cross-disease comparator analysis.

Cross-disease meta-analysis

Per-dataset patient-vs-control log2 fold-changes and standard errors for each focal gene were combined across the available cohorts using a random-effects meta-analysis (REML estimator, `metafor::rma`, v4.6-0; [17]). Heterogeneity was quantified with Cochran's *Q* and *I*². Forest plots for *IDI2-AS1* and *IL5* were generated with `metafor::forest` (Fig. 2). The Ensembl-ID gene rows for *IDI2-AS1* (ENSG00000260196) and *IL5* (ENSG00000113525) — which appear in the two Ensembl-keyed datasets (GSE246323 and GSE136825) — were collapsed onto their HGNC symbols prior to pooling so that each gene contributes at most one effect estimate per dataset. Genes matched only by HGNC symbol (the comparator lncRNAs *MIR22HG*, *GABPB1-AS1*, *OIP5-AS1* and the cytokines *IL13*, *IL6*, *TNF*, *IFNG*, *IL4*) were pooled across the three symbol-keyed datasets in which they passed the count-filter (typically *k* = 3, with *k* = 2 for *IL4*, which was below the count threshold in atopic-dermatitis skin). *IDI2-AS1* was pooled across all five datasets (*k* = 5) by virtue of being matched on both its HGNC symbol and its Ensembl ID; *IL5* was pooled across four (*k* = 4; below threshold in GSE246323).

Sample-level correlation and cell-type-adjusted analysis

Within each dataset, per-sample IDI2-AS1 and IL5 abundance was extracted from variance-stabilized counts (DESeq2::vst(blind = TRUE), raw-count datasets) or from log2(FPKM + 1) values (GSE58640). Pairwise Spearman correlation between IDI2-AS1 and IL5 was computed per dataset (Fig. 3; Table 3). To test whether any bulk-tissue

IDI2-AS1↔*IL5* covariation could be attributed to shared expression in allergic-effector cells, two compartment-specific marker-gene signatures were constructed from canonical eosinophil markers (*CLC*, *EPX*, *PRG2*, *RNASE2*, *RNASE3*, *SIGLEC8*, *CCR3*) and Th2 / type-2 markers (*GATA3*, *IL4R*, *CCR4*, *PTGDR2*, *IL13*, *IL17RB*). For each sample, a signature score was computed as the mean of the per-gene z-scores across the panel (per-dataset standardization). Cell-type-adjusted Spearman partial correlation between *IDI2-AS1* and *IL5* was then estimated with `ppcor::pcor.test` [18], conditioning on both signature scores (Fig. 4 and Supplementary Fig. 1; Table 3). For the largest cohort (GSE152004 asthma, $n = 695$), a mediation-style decomposition was performed with `mediation::mediate` (v4.5.0; [19]) using *IDI2-AS1* as treatment, the eosinophil signature score as mediator, and *IL5* as outcome (1,000 nonparametric bootstrap resamples; percentile confidence intervals). The eosinophil signature was used as the sole mediator in this decomposition — rather than including the Th2 signature in parallel — because eosinophils are the cellular endpoint of the *IL5* axis, are quantitatively the dominant *IL5*-expressing population among circulating and tissue-infiltrating leukocytes in active allergic disease, and were the compartment that most strongly attenuated the partial correlation in the asthma cohort. Because the underlying data are cross-sectional observational measurements, the resulting ACME and ADE estimates should be interpreted as a statistical decomposition under the assumed mediator–outcome model, not as identification of a causal effect in the strict counterfactual sense. Mediator and outcome models were fit by ordinary least squares (`m_model: eos_score ~ idi2as1`; `o_model: il5 ~ idi2as1 + eos_score`), and the bootstrap-based average mediated effect (ACME, in the sense of `mediation::mediate`), average direct effect (ADE), and proportion mediated were extracted from the bootstrap distribution. The marker-signature scoring approach was chosen instead of CIBERSORTx [20] for full local reproducibility from the public count matrices; CIBERSORTx-ready CPM/FPKM input matrices are nonetheless provided in `results/intermediate/cibersortx_input/` for independent verification with the LM22 reference (`code/R/08_prepare_cibersortx_input.R`).

Specificity heatmaps

To place the *IDI2-AS1* result in the context of related lncRNAs and the broader type-2 cytokine network, two heatmaps were produced with `ComplexHeatmap` (v2.18.0; [21]) and displayed as panels A (lncRNAs) and B (cytokines) of Fig. 5. For each panel, per-dataset patient-vs-control log₂ fold-changes were assembled into a gene × dataset matrix; rows were ordered with the focal gene (*IDI2-AS1* or *IL5*) on top, columns were ordered by disease, and disease and total- n annotations were drawn above the heatmap. The color scale was symmetric around zero and clipped at ± 4 log₂ units to keep the *IL13*-in-AD outlier (log₂FC ≈ 4.9) from compressing the visible dynamic range of the rest of the panel.

Code and data availability

All analysis code, intermediate result tables, and figure-generating scripts are publicly available at <https://github.com/hidenori-tani/IDI2-AS1-allergy> (commit hash recorded at submission) under an MIT license. Software environment is captured by `renv.lock` (R 4.3.2, Bioconductor 3.18). All input data are public GEO datasets cited above; no new sequencing data were generated for this study.

Results

Bulk-tissue *IDI2-AS1* expression is invariant across four IL5-driven allergic diseases

To test whether the in vitro *IDI2-AS1* → *IL5* repressive axis described in our recent T-cell-line study [5] is reflected in patient tissues, we re-analyzed five public bulk RNA-seq cohorts spanning four allergic diseases (Fig. 1; Table 1; total $n = 856$ samples). At the per-cohort level, patient-vs-control differential expression for *IDI2-AS1* was small and statistically non-significant in every dataset (Table 2): $\log_2FC$ ranged from -0.234 (atopic dermatitis, GSE121212; $p_{adj} = 0.41$) to $+0.247$ (eosinophilic esophagitis, GSE58640; $p_{adj} = 0.15$), with intermediate values in the asthma ($+0.0015$), CRSwNP (-2.8×10^{-6}), and EoE-Kleuskens (-0.036) cohorts. At the meta-analytic level, a random-effects pooling of all five datasets returned an overall $\log_2FC$ of $+5.1 \times 10^{-5}$ (95 % CI $[-0.005, +0.005]$, $p = 0.98$) with negligible heterogeneity ($I^2 = 0.26$ %, Cochran's Q $p = 0.47$; Fig. 2A). Importantly, this null is not a failure of the pipeline: in the same datasets we recovered the canonical type-2 / inflammatory signature with the expected direction and magnitude — *IL13* was strongly upregulated in atopic dermatitis ($\log_2FC = +4.92$, $p_{adj} = 2 \times 10^{-30}$) and in EoE (Sherrill, $\log_2FC = +0.91$, $p_{adj} = 2 \times 10^{-5}$); *IL6* and *TNF* were significantly elevated in AD ($\log_2FC = +3.25$ and $+0.89$; both $p_{adj} < 10^{-7}$); and *IL5* itself was upregulated in asthma ($\log_2FC = +0.42$, $p_{adj} = 0.014$) and EoE-Sherrill ($\log_2FC = +0.61$, $p_{adj} = 0.002$; see also the *IL5* random-effects forest plot in Fig. 2B, pooled $\log_2FC = +0.22$, $I^2 = 82$ %). The five-dataset $I^2 \approx 0$ % observation for *IDI2-AS1* therefore reflects a robust convergent finding — *all* cohorts independently report no bulk-tissue effect — rather than noise or pipeline artifact.

IDI2-AS1 and *IL5* covary positively at the bulk-tissue level — opposite to the cell-line prediction

We next asked whether the within-cell repressive relationship between *IDI2-AS1* and *IL5* documented in [5] is detectable as a sample-level correlation in the four cohorts where both transcripts are reliably quantified (Fig. 3; Table 3). Contrary to the cell-line prediction of a negative association, all four datasets returned a *positive* Spearman correlation between *IDI2-AS1* and *IL5*: $\rho = +0.109$ ($p = 0.004$) in the asthma cohort

(GSE152004, $n = 695$ — by far the largest and best-powered); $p = +0.489$ ($p = 0.054$) in EoE (GSE58640, $n = 16$); $p = +0.085$ ($p = 0.48$) in CRSwNP; and $p = +0.060$ ($p = 0.63$) in atopic dermatitis. The directional inversion between the published in vitro mechanism (knock-down $\rightarrow$ IL5 $\uparrow$, predicting negative in-tissue correlation) and the in vivo bulk readout (positive correlation, significant in the largest cohort) is the central observation that motivates the cell-composition analysis of the next section.

The tissue-level $IDI2\text{-}AS1 \leftrightarrow IL5$ association is mediated by eosinophil composition

A natural explanation for the inverted-direction tissue signal is that bulk RNA-seq mixes effector cells in which both $IDI2\text{-}AS1$ and $IL5$ are co-expressed (Th2 lymphocytes, ILC2s, eosinophils) with bystander tissue cells in which neither is expressed; samples with greater allergic infiltration would then carry up both transcripts in proportion, producing a composition-driven positive correlation rather than a within-cell regulatory one. To test this directly, we built per-sample eosinophil and Th2 marker-signature scores (mean z-score across 7 eosinophil markers and 6 Th2 markers; see Methods) and re-estimated the $IDI2\text{-}AS1 \leftrightarrow IL5$ relationship as a partial Spearman correlation conditioning on both signatures (Fig. 4 and [Supplementary Fig. 1](#); Table 3). Adjustment substantially attenuated the raw association in every cohort and abolished it in the asthma dataset where it had reached significance: the asthma Spearman p dropped from $+0.109$ ($p = 0.004$) to a partial p of $+0.052$ ($p = 0.17$), the EoE p dropped from $+0.489$ to $+0.425$ ($p = 0.13$), the CRSwNP p moved slightly from $+0.085$ to $+0.135$ ($p = 0.27$), and the AD p collapsed from $+0.060$ to $+0.010$ ($p = 0.94$). To formalize the contribution of the eosinophil compartment specifically, we performed a bootstrap mediation-style decomposition in the asthma cohort (1,000 simulations; $IDI2\text{-}AS1$ as treatment, eosinophil signature as mediator, $IL5$ as outcome; Table 3). The average mediated effect (ACME) through eosinophil composition was $+0.109$ (95 % CI [$+0.041$, $+0.176$], $p = 0.002$), while the average direct effect — the within-tissue residual $IDI2\text{-}AS1 \rightarrow IL5$ association after holding the eosinophil signature constant — was $ADE = +0.012$ (95 % CI [-0.075 , $+0.106$], $p = 0.76$); the proportion mediated was 0.90. In other words, ≈ 90 % of the apparent positive $IDI2\text{-}AS1 \leftrightarrow IL5$ covariation in the largest available patient cohort is statistically attributable to shared eosinophil abundance, and the residual within-tissue direct effect is indistinguishable from zero. The smaller cohorts could not confirm or refute this on their own: the atopic-dermatitis cohort returned a directionally concordant but non-significant ACME ($+0.054$, $p = 0.10$); the CRSwNP cohort returned a non-significant ACME of opposite sign (-0.028 , $p = 0.13$); and the smaller EoE cohort (GSE58640, $n = 16$) was skipped because it fell below the prespecified 20-sample mediation threshold. The mediation-style decomposition therefore rests on the asthma cohort, with the smaller cohorts directionally

compatible but underpowered. As the underlying data are cross-sectional observational measurements, the ACME / ADE values should be interpreted as a statistical decomposition of the bulk-tissue association into a composition-associated component and a residual within-tissue component, not as identification of a causal effect in the strict counterfactual sense.

Specificity: comparator lncRNAs respond to disease whereas *IDI2-AS1* does not

To test whether the flat bulk-tissue profile of *IDI2-AS1* is a genuine biological feature or a generic property of all expressed lncRNAs in these tissues, we examined the three comparator lncRNAs that passed expression filtering — *MIR22HG*, *GABPB1-AS1*, and *OIP5-AS1* — in the three datasets in which they were quantified by HGNC symbol (GSE152004 asthma, GSE121212 atopic dermatitis, GSE58640 eosinophilic esophagitis); the fourth a-priori comparator, *LITATS1*, fell below the rowSum count threshold in all of these symbol-keyed allergic-tissue datasets and was therefore not retained. The two Ensembl-keyed datasets (GSE246323, GSE136825) do not carry these lncRNAs under symbol matching and were not included for the comparators (Fig. 5A). Each of the three comparators showed clear, disease-specific differential expression. *MIR22HG* was strongly upregulated in atopic-dermatitis skin ($\log_2FC = +0.82$, $p_{adj} = 1 \times 10^{-16}$) and in EoE-Sherrill ($\log_2FC = +0.60$, $p_{adj} = 1 \times 10^{-4}$) but moved little in asthmatic nasal epithelium ($\log_2FC = -0.02$). *GABPB1-AS1* was downregulated in two of three cohorts, reaching strong significance in atopic dermatitis ($\log_2FC = -0.50$, $p_{adj} = 5 \times 10^{-3}$) and trending negative in asthma ($\log_2FC = -0.05$, $p_{adj} = 0.06$). *OIP5-AS1* was modestly but significantly downregulated in atopic dermatitis ($\log_2FC = -0.19$, $p_{adj} = 8 \times 10^{-3}$). Pooled meta-analysis ($k = 3$ datasets per gene) confirmed nontrivial cross-disease signal for the comparators: *MIR22HG* pooled $\log_2FC = +0.46$ ($I^2 = 97\%$), *GABPB1-AS1* pooled $\log_2FC = -0.10$ ($I^2 = 97\%$), *OIP5-AS1* pooled $\log_2FC = -0.03$ ($I^2 = 85\%$). Against this background, *IDI2-AS1*, which uniquely was quantified in all five datasets ($k = 5$) by virtue of also being matched on its Ensembl ID, returned the lowest pooled effect size ($+5.1 \times 10^{-5}$) and, uniquely, near-zero between-study heterogeneity ($I^2 = 0.26\%$). The flatness of *IDI2-AS1* at the bulk-tissue level is therefore a property of this lncRNA specifically, not a generic feature of low-abundance lncRNAs in allergic tissue. Taken together, this comparator panel argues that the flat *IDI2-AS1* signal is a biological property of this transcript rather than an artifact of the pipeline or of the disease cohorts.

Specificity: the type-2 cytokine network is appropriately disease-stratified

A complementary specificity check on the cytokine side confirmed that the analysis captures known disease biology in the expected pattern (Fig. 5B). *IL13* was the strongest disease-discriminating cytokine in atopic dermatitis ($\log_2FC = +4.92$, $p_{adj} =$

2×10^{-30}) and was also significantly elevated in EoE (Sherrill $\log_2FC = +0.91$, $p_{adj} = 2 \times 10^{-5}$), as expected for the canonical type-2 effector. *IL5* was significantly elevated in asthma ($+0.42$, $p_{adj} = 0.014$) and in EoE-Sherrill ($+0.61$, $p_{adj} = 0.002$), consistent with eosinophil recruitment. *IL6* and *TNF* were strongly elevated in AD only ($\log_2FC = +3.25$ and $+0.89$; $p_{adj} = 3 \times 10^{-7}$ and 6×10^{-8}), in line with that disease's broader inflammatory profile. *IFNG* was significantly downregulated in asthmatic nasal epithelium ($\log_2FC = -0.66$, $p_{adj} = 5 \times 10^{-7}$) but upregulated in atopic-dermatitis skin ($\log_2FC = +1.43$, $p_{adj} = 6 \times 10^{-3}$), reflecting the well-known difference in IFN- γ involvement between airway and cutaneous type-2 inflammation. The cytokine panel therefore behaves as expected from established immunology, providing a positive control against which the specific flatness of *ID12-AS1* in the same cohorts can be interpreted with confidence. Taken together, the comparator-lncRNA and cytokine specificity checks jointly support that the near-null bulk-tissue *ID12-AS1* signal reflects a genuine feature of this lncRNA in these tissues rather than a technical limitation of the datasets or analysis.

Discussion

What the bulk-tissue data do — and do not — say about the in vitro mechanism

The most informative feature of the present analysis is the *combination* of two results that, in isolation, would point in opposite directions. At the cohort-summary level, *ID12-AS1* differential expression is uniformly null across five independent allergic-disease datasets, with a pooled fold-change indistinguishable from zero and $I^2 \approx 0\%$ — the kind of converging null that, in a typical bulk transcriptomic catalog, would be filed as evidence against in vivo relevance. At the sample level, however, *ID12-AS1* and *IL5* covary *positively* in every cohort that can support the test, significantly so in the asthma cohort, which is the opposite of the negative direction predicted by the published in vitro repressive mechanism [5].

The cell-composition adjustment reconciles these two observations: when the eosinophil signature is held constant in the largest available cohort, $\approx 90\%$ of the apparent association is statistically attributable to shared eosinophil abundance, and the residual within-tissue direct effect is indistinguishable from zero. Read together, the three findings refine rather than refute the in vitro mechanism: the bulk-tissue null is what one would expect for a low-abundance, cell-type-restricted regulator whose per-cell signal is averaged across a predominantly bystander biopsy, and the directionally inverted bulk correlation is what one would expect when the same biopsies vary in the abundance of the cells that co-express both transcripts. The strongest evidence comes from the asthma cohort ($n = 695$); the smaller cohorts are directionally compatible but underpowered, and we treat them as such throughout.

Biological significance: why the in vitro mechanism is invisible to bulk RNA-seq

Bulk-tissue RNA-seq measures a population-weighted average of every cell type present in the biopsy. For a regulator whose function is restricted to a small effector subset — here, presumably Th2 lymphocytes, type-2 innate lymphoid cells, mast cells, and eosinophils — two distinct quantities collapse onto the same readout: (i) the per-cell expression level inside the relevant subset, and (ii) the abundance of that subset within the biopsy. In disease tissues, both can change, and they do so in opposite directions for a putative repressor of a type-2 effector cytokine: per-cell *IDI2-AS1* would be expected to fall in active disease (consistent with the cell-line knock-down $\rightarrow$ IL5 $\uparrow$ result), while the eosinophil/Th2 compartment expands. The first effect is diluted by the second, and a simple bulk mean, computed across thousands of bystander epithelial or stromal cells in which neither *IDI2-AS1* nor *IL5* is meaningfully expressed, is biased toward zero. Our cell-type-adjusted analysis dissects these two contributions explicitly: when the eosinophil signature is held constant, the residual *IDI2-AS1* $\leftrightarrow$ *IL5* relationship in the largest cohort vanishes (ADE = +0.012, $p = 0.76$). One plausible interpretation of this pattern is that the bulk signal is dominated by composition shifts and that any within-effector regulatory direction is masked, rather than contradicted, by the bulk readout. This interpretation is also consistent with the absence of *IDI2-AS1* from previously published differential-expression catalogs of these diseases: one possibility is that the granularity at which most clinical RNA-seq studies operate is simply the wrong one for this specific molecule, although alternative explanations — including the possibility that the in vitro effect does not generalize from the cell-line context — cannot be excluded by the present data alone.

A cell-composition-aware framework for in vivo discovery of cell-type-specific lncRNA regulators

IDI2-AS1 illustrates a general problem for in vivo discovery of lncRNA regulators from bulk tissue: authentic cell-type-specific regulators can be effectively invisible in standard differential-expression catalogs. The case made here for *IDI2-AS1* generalizes beyond a single transcript. lncRNAs are characteristically lower-abundance and more cell-type-restricted than protein-coding genes [1], which means the per-cell signal of an authentic regulator is more easily diluted in bulk tissue than the signal of a typical mRNA. This implies that searching for “lncRNAs that are differentially expressed in disease tissue” — the standard catalog operation in clinical transcriptomics — will systematically miss regulators that act inside small effector subsets, even when their downstream protein-coding targets (here, *IL5*, *IL13*) are themselves easily detected as differentially expressed. The *IDI2-AS1* / *IL5* contrast is therefore a useful illustration of why future discovery of in vivo lncRNA function in

immune disease should preferentially use single-cell or spatial assays, and why a null result for an interesting lncRNA in a bulk catalog should be interpreted as “not seen at this granularity” rather than “not biologically active.” Existing single-cell airway atlases [22] provide the natural follow-up substrate: testing whether *IDI2-AS1* is enriched in eosinophils, ILC2s, or Th2 cells in patient airway tissue, and whether its per-cell expression in those subsets falls in active disease, would be the directly informative experiment.

Clinical and translational context

IL5 is the validated target of three anti-eosinophil biologics (mepolizumab and reslizumab, anti-IL5; benralizumab, anti-IL5R α), approved for severe eosinophilic asthma and, in the case of mepolizumab, also for chronic rhinosinusitis with nasal polyposis [23,24]. The broader type-2 cytokine network is pharmacologically tractable in eosinophilic esophagitis as well, where for example anti-IL13 strategies have shown histologic and endoscopic improvement [25]. The clinical success of these axes is direct evidence that *IL5* — and the type-2 effector network of which it is part — is dose-limiting for the eosinophilic phenotype, which in turn implies that *upstream* regulators of *IL5*, including non-coding regulators, are mechanistically of interest. Our results do not, by themselves, establish *IDI2-AS1* as an immediate clinical biomarker — bulk-tissue expression is flat, so circulating or biopsy-derived bulk *IDI2-AS1* is unlikely to stratify patients — and they likewise do not establish it as a therapeutic target. What they do support is the prioritization of cell-type-resolved follow-up of this lncRNA in the *IL5* axis, in preference to further bulk-tissue profiling. In that framing the present manuscript serves as the *in vivo*-context companion to the earlier *in vitro* mechanistic report and as a brief against over-interpreting bulk-tissue null results when the candidate molecule is cell-type-restricted.

Limitations and future directions

Several caveats accompany this analysis. First, we relied entirely on public datasets that were not designed for lncRNA discovery; library preparation choices (poly-A selection, ribo-depletion), sequencing depth, and read length differ between cohorts and may shape the per-dataset detection of low-abundance transcripts even when an effect would otherwise be present. Related to this, our inclusion criterion required *IDI2-AS1* to be present as a non-zero feature in each retained dataset; the conclusion therefore applies specifically to publicly available cohorts in which *IDI2-AS1* is detectable at the bulk level, not to the broader claim that bulk RNA-seq always shows the lncRNA at near-null differential expression. Second, the meta-analysis pools log2FC across biologically distinct tissues (nasal epithelium, skin, esophagus, nasal polyp). Although every contributing cohort independently converged on a near-null bulk-tissue effect for *IDI2-AS1* — which is the substantive observation — pooled

effect sizes across heterogeneous tissues are necessarily summary statistics of cohort-level convergence rather than a single biological estimand, and should be read in that light. Third, the cell-type-adjustment uses a marker-gene signature score rather than a deconvolution algorithm such as CIBERSORTx. We chose this approach for full local reproducibility from the same count matrices, and we provide CIBERSORTx-formatted inputs (`results/intermediate/cibersortx_input/`) so that readers can confirm the conclusion with the LM22 reference panel; nonetheless, the marker-score approach treats the eosinophil and Th2 compartments as one-dimensional summaries and cannot resolve subtle redistribution within those compartments. Fourth, the mediation-style decomposition is well-powered only in the asthma cohort ($n = 695$); the directionally consistent but underpowered estimates in atopic dermatitis, CRSwNP, and EoE call for replication in larger samples, and — because the data are observational and cross-sectional — the ACME / ADE values describe a statistical decomposition under the assumed mediator model rather than a strict counterfactual causal effect. Fifth, the central biological interpretation — that per-cell *ID12-AS1* falls in active disease but is masked by simultaneous expansion of the *ID12-AS1*-expressing compartment — is a hypothesis derived from the bulk data, not a direct measurement; the decisive experiment is single-cell or spatial transcriptomics of patient tissue from the same diseases. Finally, the present work does not retest the in vitro *ID12-AS1* $\rightarrow$ *IL5* mechanism itself, which was established for the HuT78 / A549 systems in our prior report [5] and would benefit from independent replication in additional cell systems; the present study asks only whether the published mechanism is recoverable at the tissue scale by bulk RNA-seq, and concludes that it is not — for reasons that are themselves biologically informative. Future work should (i) re-examine *ID12-AS1* and its regulatory circuit in single-cell airway and skin atlases, (ii) test whether *ID12-AS1* knock-down in primary human Th2 cells phenocopies the cell-line *IL5* phenotype, (iii) ask — in single-cell profiling of airway or esophageal tissue from patients before and during anti-*IL5* or anti-*IL13* biologic therapy — whether per-cell *ID12-AS1* responds to therapeutic modulation of the type-2 axis, which would provide an in vivo handle on the regulatory circuit that is not accessible from cross-sectional bulk data; and (iv) only after such cell-type-resolved confirmation, evaluate whether modulation of *ID12-AS1* in defined target lineages has any biological consequence relevant to the *IL5* axis.

In summary, across five public RNA-seq cohorts ($n = 856$) covering asthma, atopic dermatitis, eosinophilic esophagitis, and chronic rhinosinusitis with nasal polyposis, the lncRNA *ID12-AS1* — recently identified as an in vitro repressor of *IL5* — shows no patient-vs-control differential expression at the bulk-tissue level (pooled $\log_2\text{FC} \approx 0$, $I^2 = 0.26\%$); every contributing cohort independently converged on a near-null bulk-tissue effect. Sample-level *ID12-AS1* abundance instead covaries *positively* with *IL5*, and in the largest cohort (asthma, $n = 695$) a mediation-style decomposition

attributes $\approx 90\%$ of that covariation to shared expression in the eosinophil compartment, leaving no detectable within-tissue direct effect (ADE $p = 0.76$). Bulk patient transcriptomes therefore do not recover the in vitro *ID12-AS1* $\rightarrow$ *IL5* repressive direction; the pattern is most consistent with masking by eosinophil-driven shifts in tissue cell composition, with the strongest evidence coming from the asthma cohort. Whether the in vitro mechanism actually operates in patient cells — or whether the in vitro repression simply does not generalize beyond the cell-line context — cannot be adjudicated by bulk re-analysis alone, and will require single-cell or spatial transcriptomics of the same diseases, which represent the most informative next experimental step. Beyond the *ID12-AS1* case, the cell-composition-adjusted re-analysis pipeline used here is released as open code (Data Availability) and is directly applicable to other low-abundance, cell-type-restricted lncRNA candidates whose bulk-tissue null results might otherwise be mis-read as evidence against in vivo relevance.

Author contributions

Hidenori Tani (ORCID: 0000-0001-6390-4136): Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Conflict of interest

The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Funding

No specific funding was received for this work.

Acknowledgments

The author thanks the contributors of the public RNA-seq cohorts re-analyzed in this study (GSE152004, GSE121212, GSE246323, GSE58640, GSE136825) for making their data openly available, and the developers of DESeq2, apegglm, limma, metafor, ppcor, mediation, ComplexHeatmap, and CIBERSORTx for the open-source tools that made this analysis possible.

Data availability

All datasets re-analyzed in this study are publicly available from the NCBI Gene Expression Omnibus (GEO) under accession numbers GSE152004 (asthma; [7]),

GSE121212 (atopic dermatitis; [8]), GSE58640 (eosinophilic esophagitis; [9]), GSE246323 (eosinophilic esophagitis; [10]), and GSE136825 (chronic rhinosinusitis with nasal polyposis; [11]). All analysis code, intermediate result tables, and figure-generating scripts are publicly available at <https://github.com/hidenori-tani/IDI2-AS1-allergy> under an MIT license. The R software environment is captured by `renv.lock` for full reproducibility.

Generative AI disclosure

During the preparation of this manuscript, the author used Anthropic Claude Code (Claude Opus 4.7) as a writing and coding assistant to help draft and refine analysis scripts, figure-generation code, and manuscript text. All scientific content, analytical decisions, and final wording were reviewed, edited, and verified by the author, who takes full responsibility for the content of the publication.

Figure legends

Fig. 1. Study design and analytical workflow. Single-panel schematic organized as four vertical stages. **Stage 1 (Datasets):** the five public bulk RNA-seq cohorts re-analyzed in this study, spanning four IL5-driven allergic diseases — asthma (GSE152004, $n = 695$, nasal epithelium), atopic dermatitis (GSE121212, $n = 65$, skin biopsy), eosinophilic esophagitis (GSE246323, $n = 10$ and GSE58640, $n = 16$, esophageal biopsy), and chronic rhinosinusitis with nasal polyposis (GSE136825, $n = 70$, nasal polyp). **Stage 2 (Uniform pipeline):** the five sequential analytical steps applied to every dataset — (i) per-dataset differential expression (DESeq2 + apeglm for raw counts, limma-trend for FPKM), (ii) random-effects meta-analysis across cohorts (metafor: :rma, REML), (iii) sample-level Spearman correlation between *IDI2-AS1* and *IL5*, (iv) cell-type-adjusted partial correlation conditioning on eosinophil and Th2 marker-gene signature scores, and (v) bootstrap mediation-style decomposition in the asthma cohort (1 000 resamples). **Stage 3 (Three in vivo questions):** Q1 — is *IDI2-AS1* itself differentially expressed in disease tissue? Q2 — do *IDI2-AS1* and *IL5* covary at the sample level in the predicted (negative) direction? Q3 — if the bulk-tissue signal departs from the in vitro prediction, can it be ascribed to cell-composition shifts? **Stage 4 (Principal finding):** bulk-tissue *IDI2-AS1* is invariant in every cohort (pooled $\log_2FC \approx 0$; $I^2 = 0.26\%$); sample-level *IDI2-AS1* $\leftrightarrow$ *IL5* is positive in every testable cohort (opposite to the in vitro prediction); in the asthma cohort, $\approx 90\%$ of that positive association is attributed to the eosinophil compartment (ACME = $+0.109$, $p = 0.002$), with a within-tissue direct effect indistinguishable from zero (ADE = $+0.012$, $p = 0.76$). The schematic makes the central interpretive arc visible at a glance: bulk DE $\rightarrow$ bulk correlation $\rightarrow$ cell-composition adjustment $\rightarrow$ within-tissue residual.

Fig. 2. *IDI2-AS1* and *IL5* random-effects meta-analysis across five allergic-disease cohorts. (A) Forest plot for *IDI2-AS1* ($k = 5$ datasets). Per-dataset DESeq2 + apeglm or limma-trend log2FC estimates with 95 % confidence intervals are shown for each cohort, with the random-effects pooled estimate (REML, metafor: rma) at the bottom. The pooled estimate is $+5.1 \times 10^{-5}$ (95 % CI $[-0.005, +0.005]$; $p = 0.98$), with negligible between-study heterogeneity ($I^2 = 0.26$ %; Cochran's Q $p = 0.47$), indicating that all five datasets independently agree on the absence of a bulk-tissue effect. (B) Forest plot for *IL5* ($k = 4$; *IL5* was below the rowSum ≥ 10 count threshold in GSE246323 and is therefore not included in this meta-analysis). Pooled log2FC = $+0.22$ (95 % CI $[-0.15, +0.58]$; $p = 0.25$), with substantial heterogeneity driven by the strong asthma and EoE-Sherrill effects ($I^2 = 82$ %).

Fig. 3. Sample-level *IDI2-AS1* $\leftrightarrow$ *IL5* covariation per cohort. Per-sample *IDI2-AS1* vs *IL5* abundance (variance-stabilized counts for raw-count datasets, log2(FPKM + 1) for GSE58640) in the four cohorts where both transcripts are reliably quantified. Each panel is one cohort, with patients and controls overlaid (colour key: control vs patient, shown at the bottom of the figure). Spearman correlations: asthma $\rho = +0.109$, $p = 0.004$ ($n = 695$); EoE (Sherrill) $\rho = +0.489$, $p = 0.054$ ($n = 16$); CRSwNP $\rho = +0.085$, $p = 0.48$ ($n = 70$); atopic dermatitis $\rho = +0.060$, $p = 0.63$ ($n = 65$). All four cohorts show positive covariation — the opposite of the negative direction predicted by the in vitro repressive mechanism reported in [5].

Fig. 4. Cell-type adjustment dissolves the asthma *IDI2-AS1* $\leftrightarrow$ *IL5* association. (Top row) Raw sample-level *IDI2-AS1* (x-axis) vs *IL5* (y-axis) scatter, one panel per cohort, at uniform aspect ratio for direct visual comparison with the bottom rows (same data as Fig. 3). (Middle row, “Eosinophil signature”) *IDI2-AS1* (x-axis) vs the eosinophil marker-signature score (y-axis, mean z-score across CLC, EPX, PRG2, RNASE2, RNASE3, SIGLEC8, CCR3). (Bottom row, “Th2 / type-2 signature”) *IDI2-AS1* vs the Th2 / type-2 marker-signature score (mean z-score across GATA3, IL4R, CCR4, PTGDR2, IL13, IL17RB). Signatures are standardized within each cohort. The strong positive association between *IDI2-AS1* and the eosinophil signature in the asthma cohort, mirrored by the *IDI2-AS1* $\leftrightarrow$ *IL5* association in the same cohort (top row), is the visual basis for the cell-composition explanation formally quantified as a partial Spearman correlation and a bootstrap-mediation-style decomposition in Table 3. [Fig. 4. Cell-type adjustment dissolves the asthma *IDI2-AS1* \$\leftrightarrow\$ *IL5* association. \(A\) Raw sample-level *IDI2-AS1* \(x-axis\) vs *IL5* \(y-axis\) scatter, one panel per cohort \(same data as Fig. 3, shown here at uniform aspect ratio for direct visual comparison with panel B\). \(B\) *IDI2-AS1* \(x-axis\) vs the eosinophil marker-signature score \(y-axis, mean z-score across CLC, EPX, PRG2, RNASE2, RNASE3, SIGLEC8, CCR3\), one panel per cohort. Signatures are standardized within each cohort. The strong positive association between *IDI2-AS1* and the eosinophil signature in the asthma cohort, mirrored by the *IDI2-AS1*](#)

↔ IL5 association in the same cohort (panel A), is the visual basis for the cell-composition explanation formally quantified as a partial Spearman correlation and a bootstrap mediation-style decomposition in Table 3. The complementary IDI2-AS1 vs Th2 / type-2 marker-signature scatter is shown as Supplementary Figure 1.

Fig. 5. Specificity panels — comparator lncRNAs and type-2 cytokines across the disease cohorts. **(A)** Heatmap of patient-vs-control log2 fold-changes for *IDI2-AS1* and three comparator lncRNAs (*MIR22HG*, *GABPB1-AS1*, *OIP5-AS1*) across all five cohorts. Rows: lncRNAs, anchored on *IDI2-AS1*. Columns: cohorts, with per-cohort total sample size annotated above. Colour scale: symmetric around zero (blue = downregulated in patients; red = upregulated), clipped at ± 1 log2 unit (which spans the full observed range for these lncRNAs). Comparator lncRNAs show clear disease-specific differential expression (e.g. *MIR22HG* +0.82 in atopic dermatitis, *GABPB1-AS1* -0.50 in atopic dermatitis), whereas *IDI2-AS1* is uniquely flat in every cohort. The three comparator lncRNAs are matched by HGNC symbol only and therefore appear as empty cells in the two Ensembl-keyed cohorts (GSE246323 and GSE136825); the a-priori fourth comparator *LITATS1* fell below the rowSum ≥ 10 count threshold in every symbol-keyed allergic-tissue dataset and is therefore not displayed. *IDI2-AS1* (matched on both HGNC symbol and Ensembl ID) is uniquely quantified in all five cohorts. **(B)** Heatmap of patient-vs-control log2 fold-changes for six type-2 / inflammatory cytokines (*IL5*, *IL4*, *IL13*, *IL6*, *TNF*, *IFNG*) across four cohorts (GSE152004, GSE121212, GSE58640, GSE136825; GSE246323 is not shown because these cytokine symbols were not recoverable from its Ensembl-keyed matrix). Colour conventions are the same as in panel A, but the scale is clipped at ± 4 log2 units to accommodate the *IL13*-in-AD outlier (+4.92). *IL13* is strongly elevated in atopic dermatitis (log2FC = +4.92, padj = 2×10^{-30}) and in EoE-Sherrill (+0.91, padj = 2×10^{-5}); *IL5* is elevated in asthma (+0.42, padj = 0.014) and in EoE-Sherrill (+0.61, padj = 0.002); *IL6* and *TNF* are strongly elevated in atopic dermatitis only; *IFNG* is downregulated in asthmatic nasal epithelium (-0.66, padj = 5×10^{-7}) but upregulated in atopic-dermatitis skin (+1.43). The cytokine panel behaves as expected from established type-2 immunology, providing a positive control against which the specific flatness of *IDI2-AS1* in the same cohorts can be interpreted.

Supplementary Figure legends

Supplementary Fig. 1. IDI2-AS1 vs Th2 / type-2 marker signature per cohort. Per-sample IDI2-AS1 (x-axis) vs Th2 / type-2 marker-signature score (y-axis, mean z-score across GATA3, IL4R, CCR4, PTGDR2, IL13, IL17RB) in each of the five cohorts. Signatures are standardized within each cohort. This panel complements Fig. 4B (eosinophil signature) and shows the cohort-by-cohort relationship between IDI2-AS1 and the broader type-2 effector compartment.

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